CT PERFORMANCE EVALUATION REPORT

PRE-TEST / POST FAN CHANGE / POST DISTRIBUTION CHANGE

— THREE-STAGE SINGLE CELL ASSESSMENT —

COOLING TOWER — CELL 2 | 3-FAN INDUCED DRAFT

OWNER / CLIENT	DUMMY INDUSTRIES LTD.
TESTING AGENCY	SS COOLING TOWER CONSULTANTS (SSCTC)
DATE OF TEST	12-Mar-2026
REPORT DATE	19-Apr-2026
ASSESSMENT METHOD	CTI ATC-105 Merkel Method — Applied independently to each of 3 test stages

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PREAMBLE

This report presents the CT performance evaluation for Cell 2. Three stages: Pre-Test, Post Fan Change, and Post Distribution Change, each assessed using CTI ATC-105 Merkel Method.

Field measurements were recorded by SSCTC engineers under steady-state conditions. All instruments were calibrated prior to testing.

MEMBERS PRESENT FOR THE TEST

CLIENT

- Mr. A. Sharma — Plant Manager
- Mr. B. Patel — Maintenance Engineer

SS COOLING TOWER CONSULTANTS (SSCTC)

- Er. R. Mehta — Lead Test Engineer
- Er. S. Kapoor — Field Engineer

ASSESSMENT METHOD & ANALYSIS

The CTI ATC-105 Merkel method computes tower characteristic (KaV/L) from field data and compares to design demand.

Three independent stages were conducted. Cross Plot 1 gives CWT vs Range at Test WBT; Cross Plot 2 gives Flow vs CWT at Design conditions.

INSTRUMENT PLACEMENT AND READINGS

Wet Bulb: Aspirated psychrometer at air inlet face. Three readings at 15-minute intervals.

HWT: Calibrated RTD in inlet pipe. CWT: Sensors in basin sump.

CONCLUSION AND REMARKS

1. Cell 2 was below design across all three test stages, with the largest shortfall in Test 1.
2. Post fan-blade pitch adjustment (Test 2) improved performance by 0.42 C, reducing shortfall from 1.85 C to 1.43 C.
3. Post distribution nozzle replacement (Test 3) gave a further 0.38 C improvement, yielding a final shortfall of 1.05 C.

SUGGESTIONS AND SPECIAL NOTES

1. Fan blade pitch should be re-verified every 6 months.
2. Distribution header and nozzles should be inspected and cleaned annually.
3. A follow-up thermal test is recommended after 12 months of full operation.

FINAL DATA AND CONCLUSION OF PRE AND POST TEST

The following table presents a consolidated comparison of all measured and calculated parameters across the three test stages conducted during this evaluation.

Parameter	Unit	TEST 1 Pre-Test	TEST 2 Post Fan Change	TEST 3 Post Distr.
Water Flow Rate	m3/hr	850.00	853.50	856.20
HWT	C	42.10	42.30	42.20
CWT (Test)	C	29.85	29.63	29.45
CWT Shortfall	C	-1.85	-1.43	-1.05
Capability (%)	%	108.6	106.4	104.7

NOTES

1. All temps in Celsius. Flow in m3/hr.

ATC-105 PERFORMANCE EVALUATION — TEST 1 — PRE TEST

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	900.00	850.00
Hot Water Temp	°C	40.00	42.10
Cold Water Temp	°C	28.00	29.85
Wet Bulb Temp	°C	24.00	24.50
Range	°C	12.00	12.25
Approach	°C	4.00	5.35

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	9.6 °C	26.80	25.60	24.60
100%	12.0 °C	28.90	27.70	26.50
120%	14.4 °C	30.80	29.50	28.30

STEP 2 — CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

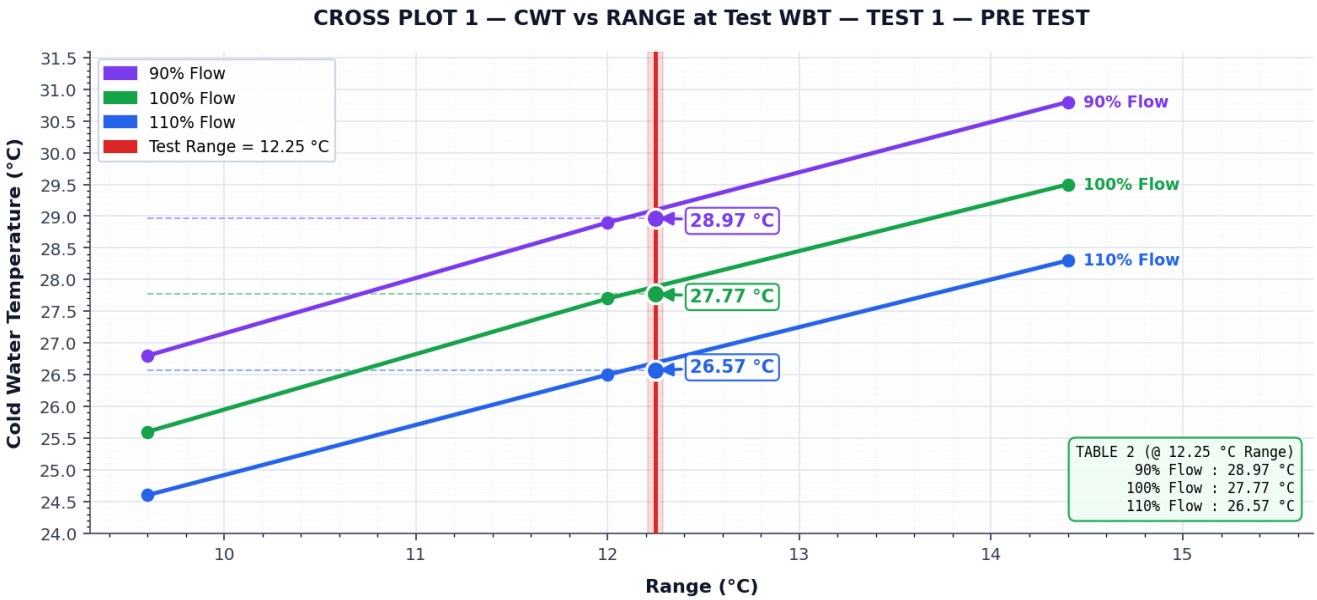

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 1 — PRE TEST

TABLE 2 — CWT at Test Range = 12.25 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 12.25°C
90% Flow	810.00	28.97
100% Flow	900.00	27.77
110% Flow	990.00	26.57

STEP 3 ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**850.00 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**28.97 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**27.77 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**26.57 °C**
vi. Test Hot Water Temperature (HWT)	**42.10 °C**
vii. Test Cold Water Temperature (CWT)	**29.85 °C**
viii. Test Wet Bulb Temperature (WBT)	**24.50 °C**
Q_adj — Adjusted Water Flow (Density Corrected)	**847.20 m³/hr**

STEP 4 CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow where the curve meets the Design CWT — this is the Predicted Flow used in Step 5.

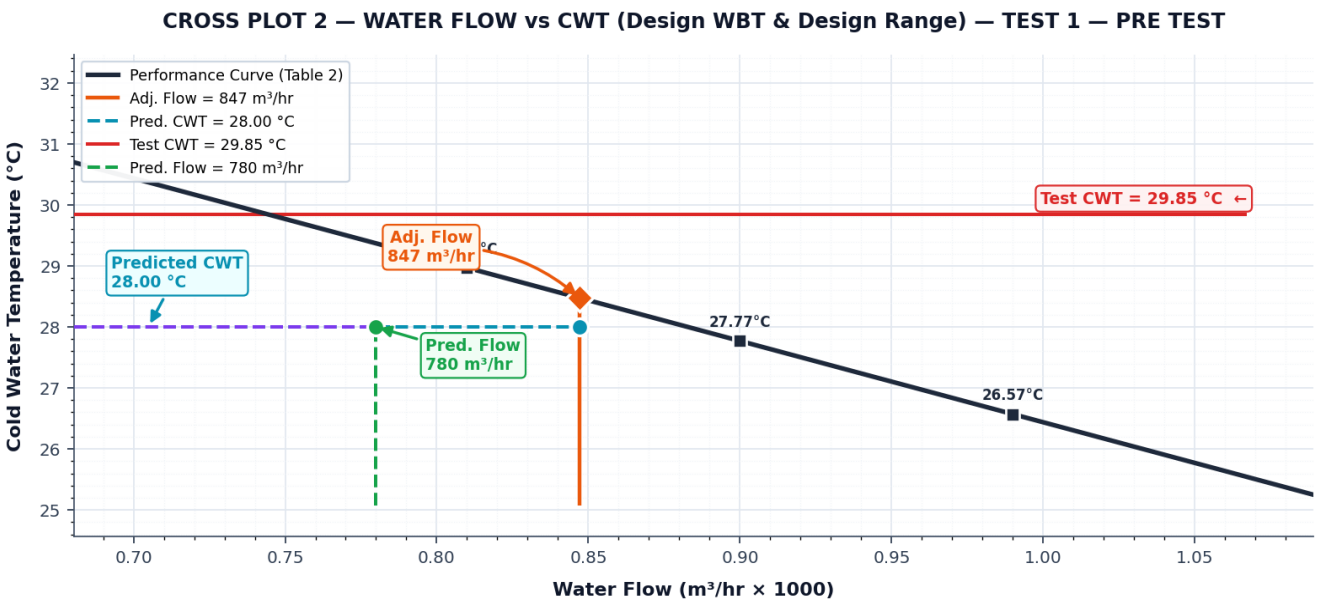

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 1 — PRE TEST

FINAL ATC-105 RESULTS — TEST 1 — PRE TEST

STEP 5 FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	847.20 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	28.00 °C
iii	Test CWT — Recorded during test	29.85 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	780.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-1.85 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	108.60 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 1 — PRE TEST

CWT SHORTFALL IN PERFORMANCE

-1.85 °C

Test CWT (29.85°C) vs Predicted CWT (28.00°C)

COOLING TOWER CAPABILITY

108.6 %

= 847 / 780 × 100

Tower is **BELOW specification** by 1.85°C — Capability: **108.6%**

ATC-105 PERFORMANCE EVALUATION — TEST 2 — POST FAN CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	900.00	853.50
Hot Water Temp	°C	40.00	42.30
Cold Water Temp	°C	28.00	29.63
Wet Bulb Temp	°C	24.00	24.70
Range	°C	12.00	12.07
Approach	°C	4.00	4.93

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	9.6 °C	27.00	25.80	24.80
100%	12.0 °C	29.10	27.90	26.70
120%	14.4 °C	30.90	29.70	28.40

STEP 2 CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

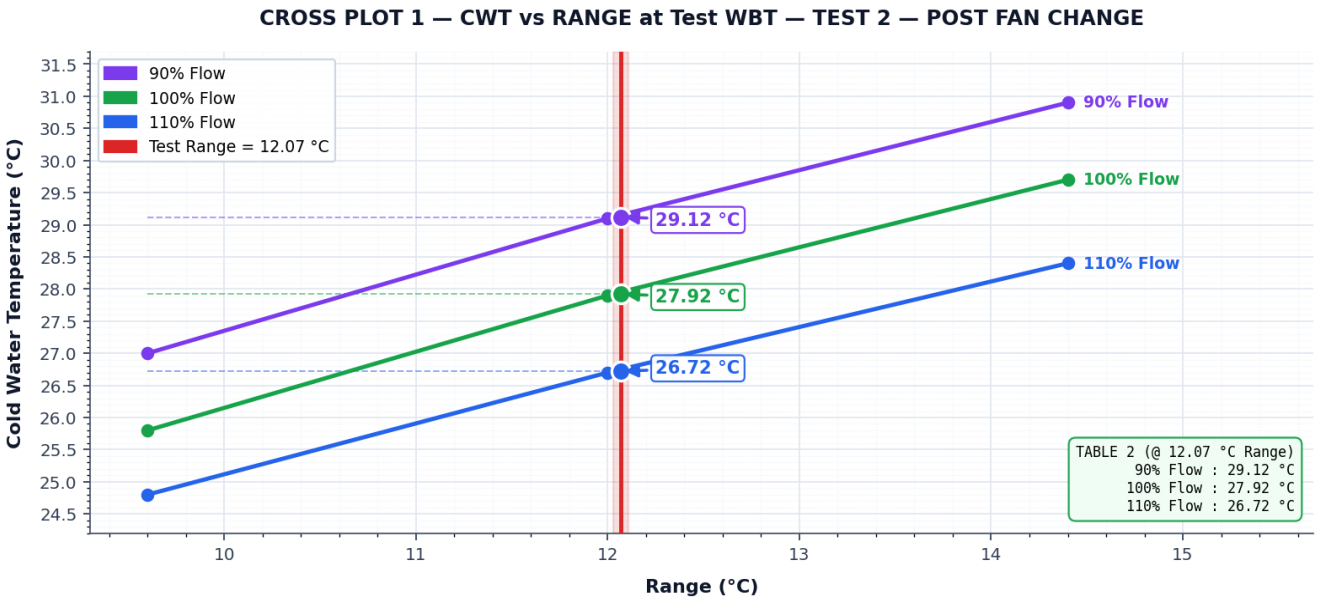

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 2 — POST FAN CHANGE

TABLE 2 — CWT at Test Range = 12.07 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 12.07°C
90% Flow	810.00	29.12
100% Flow	900.00	27.92
110% Flow	990.00	26.72

STEP 3 ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**853.50 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**29.12 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**27.92 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**26.72 °C**
vi. Test Hot Water Temperature (HWT)	**42.30 °C**
vii. Test Cold Water Temperature (CWT)	**29.63 °C**
viii. Test Wet Bulb Temperature (WBT)	**24.70 °C**
Q_adj — Adjusted Water Flow (Density Corrected)	**850.80 m³/hr**

STEP 4 CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow where the curve meets the Design CWT — this is the Predicted Flow used in Step 5.

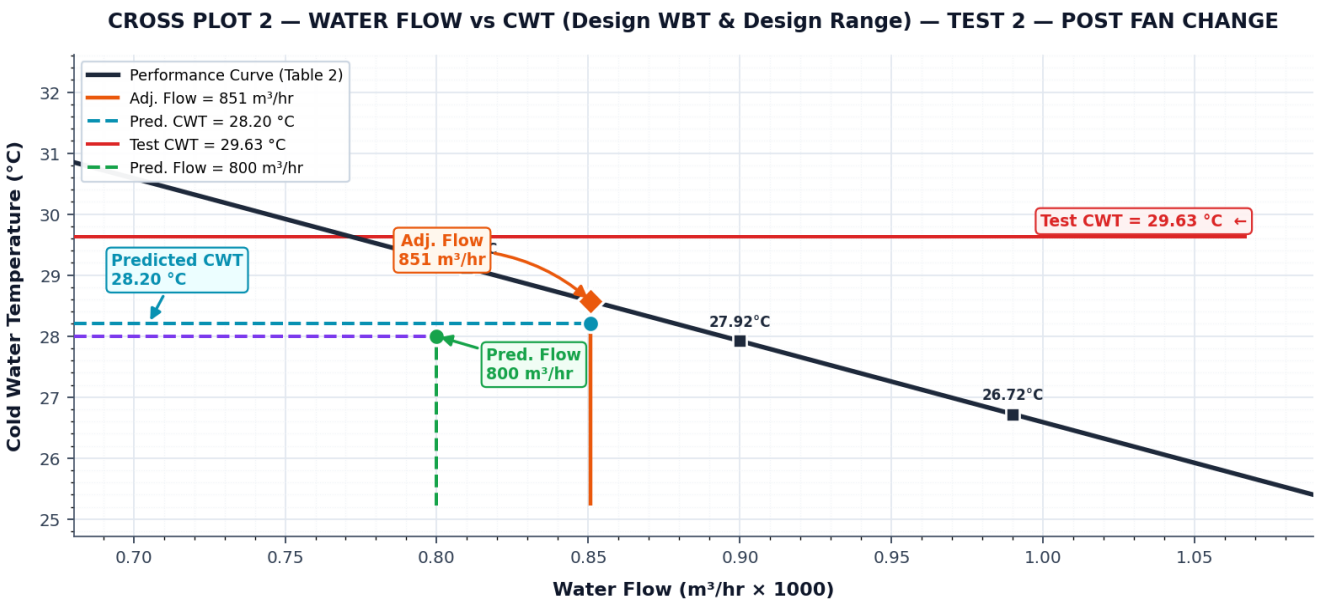

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 2 — POST FAN CHANGE

FINAL ATC-105 RESULTS — TEST 2 — POST FAN CHANGE

STEP 5 FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	850.80 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	28.20 °C
iii	Test CWT — Recorded during test	29.63 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	800.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-1.43 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	106.40 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 2 — POST FAN CHANGE

CWT SHORTFALL IN PERFORMANCE

-1.43 °C

Test CWT (29.63°C) vs Predicted CWT (28.20°C)

COOLING TOWER CAPABILITY

106.4 %

= 851 / 800 × 100

Tower is **BELOW specification** by 1.43°C — Capability: **106.4%**

ATC-105 PERFORMANCE EVALUATION — TEST 3 — POST DISTRIBUTION CHANGE

DESIGN VS RECORDED TEST CONDITIONS

Parameter	Unit	Design Conditions	Recorded Test Conditions
Water Flow	m³/hr	900.00	856.20
Hot Water Temp	°C	40.00	42.20
Cold Water Temp	°C	28.00	29.45
Wet Bulb Temp	°C	24.00	24.60
Range	°C	12.00	12.25
Approach	°C	4.00	4.85

STEP 1

TABLE 1 — CWT vs Range at Test WBT

Using the Merkel method, predicted Cold Water Temperatures (CWT) are computed at three fan speeds (90%, 100%, 110% of design flow) for three range values (80%, 100%, 120% of design range) at the recorded Test Wet Bulb Temperature.

Range %	Range (°C)	CWT @ 90% Flow	CWT @ 100% Flow	CWT @ 110% Flow
80%	9.6 °C	27.20	26.00	25.00
100%	12.0 °C	29.30	28.10	26.90
120%	14.4 °C	31.10	29.90	28.60

STEP 2 — CROSS PLOT 1 — CWT vs Range at Test WBT

Plot the three CWT lines (90%, 100%, 110% flow) from Table 1 against Range. Draw a vertical line at the Test Range to read off the CWT for each flow condition. These three (Flow, CWT) pairs form Table 2.

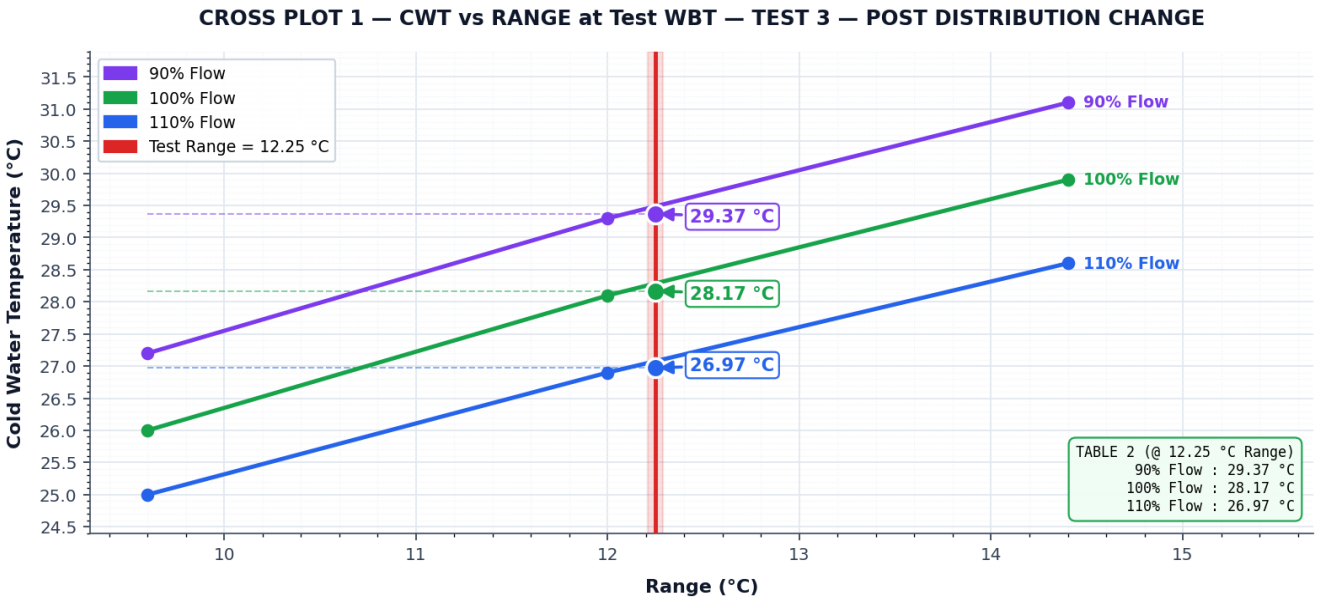

Figure 1: Cross Plot 1 — CWT vs Range at Test WBT — TEST 3 — POST DISTRIBUTION CHANGE

TABLE 2 — CWT at Test Range = 12.25 °C

Flow Condition	Water Flow (m³/hr)	CWT at Range 12.25°C
90% Flow	810.00	29.37
100% Flow	900.00	28.17
110% Flow	990.00	26.97

STEP 3 ADJUSTED WATER FLOW — Air Density Correction

Adjust the recorded water flow for air density differences between test and design conditions: Q_adj = Q_test × (Air Density Ratio)^0.5. The density ratio accounts for atmospheric variation in air mass flow delivered by the fan.

i. Recorded Test Water Flow (Q_test)	**856.20 m³/hr**
ii. Air Density Ratio (ρ_design / ρ_test)	**1.00**
iii. CWT @ 90% Design Flow (Cross Plot 1)	**29.37 °C**
iv. CWT @ 100% Design Flow (Cross Plot 1)	**28.17 °C**
v. CWT @ 110% Design Flow (Cross Plot 1)	**26.97 °C**
vi. Test Hot Water Temperature (HWT)	**42.20 °C**
vii. Test Cold Water Temperature (CWT)	**29.45 °C**
viii. Test Wet Bulb Temperature (WBT)	**24.60 °C**
Q_adj — Adjusted Water Flow (Density Corrected)	**853.40 m³/hr**

STEP 4 CROSS PLOT 2 — Water Flow vs CWT (Design WBT & Range)

Plot (Water Flow, CWT) pairs from Table 2. Locate Q_adj on the performance curve to read the Predicted CWT. Then find the water flow where the curve meets the Design CWT — this is the Predicted Flow used in Step 5.

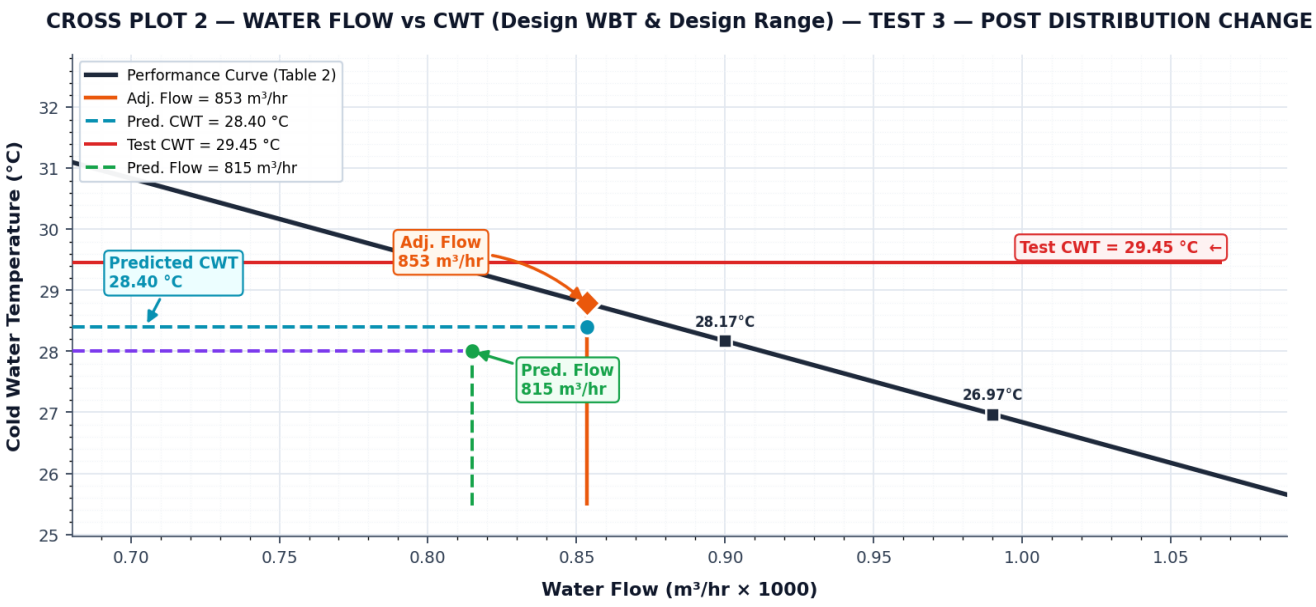

Figure 2: Cross Plot 2 — Water Flow vs CWT (Design WBT & Range) — TEST 3 — POST DISTRIBUTION CHANGE

FINAL ATC-105 RESULTS — TEST 3 — POST DISTRIBUTION CHANGE

STEP 5 FINAL EVALUATION — CWT Shortfall & Capability

Compare the Predicted CWT (from Cross Plot 2 at Q_adj) with the Test CWT. CWT Shortfall = Test CWT – Predicted CWT. A negative shortfall means the tower performed below specification. Capability % = (Q_adj / Predicted Flow) × 100.

Step	Parameter	Value
i	Q_adj — Adjusted Water Flow (density corrected)	853.40 m³/hr
ii	Predicted CWT — from Cross Plot 2 at Q_adj	28.40 °C
iii	Test CWT — Recorded during test	29.45 °C
iv	Predicted Water Flow — at Design CWT on Cross Plot 2	815.00 m³/hr
v	CWT Shortfall = Test CWT – Predicted CWT	-1.05 °C
vi	Cooling Tower Capability = (Q_adj / Pred. Flow) × 100	104.70 %
vii	Air Density Ratio applied	1.00

CONCLUSION — TEST 3 — POST DISTRIBUTION CHANGE

CWT SHORTFALL IN PERFORMANCE

-1.05 °C

Test CWT (29.45°C) vs Predicted CWT (28.40°C)

COOLING TOWER CAPABILITY

104.7 %

= 853 / 815 × 100

Tower is **BELOW specification** by 1.05°C — Capability: **104.7%**

——— END OF REPORT ———

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